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The mediating role of presence differs across types of spatial learning in immersive technologies

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ABSTRACT

The effects of immersive technology on learning have been mixed. It is therefore important to determine the factors that affect when and why immersive technologies are and are not effective. One psychological construct proposed to explain why higher levels of immersive technology may lead to better learning compared to lower immersion is presence, or the subjective feeling of "being there." Participants completed a spatial task in three levels of immersive technology, reported the amount of presence felt, and completed learning outcome tasks measuring three levels of spatial knowledge: landmark, route, and survey knowledge. The relationships between the level of immersive technology, presence, and spatial learning outcomes were examined. The highest immersion condition led to better performance on landmark, route, survey, and overall spatial knowledge, and also led to higher levels of presence. Higher presence led to better performance on route, survey, and overall spatial knowledge. However, presence only significantly mediated the relationship for survey knowledge, and effects of low vs. medium immersion condition on learning and presence often did not differ, despite the devices having largely different affordances. The relationship between immersion and learning is thus complex, depends on type of learning, and may be mediated by both presence and non-presence effects on cognitive load.

1. Introduction

In training and teaching contexts, using Virtual Reality (VR) to deliver information to a learner is valuable because it can lessen the burden on in-person instructors and reduce the need for physical supplies and equipment. The flexibility of VR is also important. VR can simulate any location with any number of pedagogical agents, allowing learners to experience virtual simulations of real-world scenarios that may be dangerous or expensive, or scenarios that would otherwise be unfeasible in the real world. VR has been used in a variety of academic domains and fields. Examples include learning laboratory techniques for a chemistry experiment (Makransky, Terkildsen, & Mayer, 2017), exploring the inside of an animal cell (Parong & Mayer, 2018), training in emergency situations (Buttussi & Chittaro, 2018; Carlson & Caporusso, 2019; Shu, Huang, Chang, & Chen, 2018), learning medical procedures such as dissection (Aggarwal et al., 2019; Moro, Štromberga, Raikos, & Stirling, 2017; Riva, 2002), and acquiring spatial knowledge of an environment (Waller, Hunt, & Knapp, 1998).

1.1. VR immersion and learning

VR technologies vary in their levels of *immersion*, which is defined as an objective property of the technology and its affordances (Cummings & Bailenson, 2016). VR systems considered higher in immersion (e.g., head-mounted displays, HMDs) often include higher resolution head tracking, faster update rates, stereoscopic vision (rather than monoscopic vision), greater degrees of freedom of head rotation and tracking, higher image and sound quality, more external visual occlusion, and/or a larger field of view compared to VR systems considered lower in immersion (e.g., desktop monitor; Cummings & Bailenson, 2016). Along with the apparent practical benefits of VR, there is a popular underlying notion that highly immersive technology is the most beneficial for

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learning; lab studies and consumer surveys report that students and teachers believe using more-immersive VR in the classroom would improve learning outcomes (Hew & Cheung, 2010; Makransky et al., 2017; Samsung Business USA, 2016). However, research has shown mixed results. Some research has shown that technologies with higher immersion lead to better learning outcomes compared to technologies with lower immersion (e.g., Kozhevnikov, Gurlitt, & Kozhevnikov, 2013; Markowitz, Laha, Perone, Pea, & Bailenson, 2018; Webster, 2016), but sometimes learning outcomes do not differ (e.g., Aggarwal, et al., 2019; Moreno & Mayer, 2004). In other cases, higher immersion leads to poorer learning outcomes (e.g., Makransky et al., 2017; Parong & Mayer, 2018). These mixed findings may become clearer if potentially mediating psychological states are considered. We therefore aimed to uncover mediating factors that can explain when and why higher immersion learning environments are beneficial. By doing so, future training can then be better designed to maximize learning outcomes.

1.1.1. Immersion and presence

In instances where higher immersion boosts learning performance, one possible explanation is that immersion makes the learners feel like they are really in the virtual environment; in other words, they are experiencing *presence*. Presence can be defined as the psychological experience of "being there," or the experience of being in one environment even when the person is physically in an another (Cummings & Bailenson, 2016; Witmer & Singer, 1998). Wirth and colleagues (Wirth et al., 2007) describe a two-step process by which presence is experienced. First, the user uses spatial cues to perceive that the virtual environment is a plausible space. Second, the user then also experiences him or herself being located in the space with perceived possibilities to act. The virtual environment is more likely to be perceived as a plausible space that can be acted upon if visual and auditory cues are rich in quality and logically consistent with the real world. For example, some factors that may play a role in perceived presence within these two steps include (1) control factors, such as degree and immediacy of control, (2) sensory factors, such as environmental richness and multimodal presentation, (3) distraction factors, such as isolation and interference awareness, and (4) realism factors, such as information consistency with the objective world (Witmer & Singer, 1998). Thus, from this rationale, learning environments with higher immersive qualities may elicit greater psychological presence.

Presence is typically measured using self-report tools, with users retroactively reporting how much they felt like they were really in the virtual environment based on questions about sensory information, involvement, interactions, interface, etc. (Usoh, Catena, Arman, & Slater, 2000; Witmer & Singer, 1998). Technologies with higher immersion have been shown to increase a user's presence in laboratory studies (e.g., Makransky et al., 2017; Moreno & Mayer 2004), and a meta-analysis reports that across different levels of immersive technologies, immersion has a medium-sized effect on presence (Cummings & Bailenson, 2016).

1.1.2. Presence and learning

Previous research has found presence to be associated with learning outcomes (Makransky & Lilleholt, 2018; Schrader & Bastiaens, 2012). One theory that could explain why presence in a virtual environment increases learning is the cognitive theory of multimedia learning (CTML; Mayer, 2014), which is derived from the closely related cognitive load theory (CLT; Chandler & Sweller, 1991; Sweller, 1994). These theories propose that learners have limited cognitive resources during information acquisition and that learning depends on using the available cognitive resources efficiently. CTML specifies three types of cognitive load: essential, generative, and extraneous. Essential load is the cognitive load induced by the to-be-learned information and depends on the level of complexity of the information and the learner's prior knowledge. Generative load is the cognitive load required to deeply process the incoming stimuli in working memory and effectively store it in long term memory. Doing so requires the selection of relevant information, the organization of a coherent mental model of the information, and integration of this model with prior knowledge in long term memory. Extraneous load is the cognitive load that is associated with information that is not essential to the lesson, such as irrelevant verbal or pictorial information within the lesson or distractions from outside of the lesson. Successful learning occurs when extraneous load is minimized, which leaves more cognitive resources to devote to essential and generative load (Mayer, 2009, 2014).

Presence in a virtual learning environment may help attenuate extraneous load, manage cognitive load, and foster efficient generative load. Feeling present in a virtual environment may indicate reduced perception of the existence of the physical interface. For example, this would diminish the distraction and extraneous load resulting from wearing an HMD or interacting with the controller (Schrader & Bastiaens, 2012). Thus, more cognitive resources could be allocated to processing the essential information. Presence may also help manage essential load in a spatial learning task. In a task that requires understanding the relationships between one's position and locations in the environment, heightened presence reduces the essential load of translating one's position from the physical environment to the virtual environment. Additionally, presence could foster more efficient generative processing by creating more salient imagery and richer episodic memories of the environment, which can then be more easily organized in working memory and recalled from long term memory. Krokos, Plaisant, and Varshney (2019) examined the use of virtual reality as a proxy for a "memory palace." A memory palace is a spatial imagery mnemonic used to assist recall by associating to-be-remembered information with specific features of an imagined environment, such as the people, objects, or rooms therein. Participants were asked to remember a set of faces in a virtual memory palace, and those who used an HMD had significantly better recall than those who used a desktop monitor. The researchers propose that the experience of presence through immersion could play a role in enhancing retention and recall. In related research in a non-virtual reality setting, a higher sense of presence while watching a movie was associated with better factual memory of events of the movie (Makowski, Sperduti, Nicolas, & Piolino, 2017).

From the above theoretical background and findings, one might expect a straightforward relationship between immersion and presence and then between presence and learning, leading perhaps to an overall consistent relationship between immersion and learning. However, the literature shows these individual relationships are not so straightforward. For example, some studies showed that although learners reported higher presence in a highly immersive learning environment, they had equal or worse performance on learning outcome tasks compared to learners in lower immersion learning environments (e.g., Makransky et al., 2017; Moreno & Mayer, 2004). However, these were not tasks in the spatial domain. Presence may be more critical for spatially related tasks. Similarly, different types of learning tasks within a domain may affect the extent to which presence influences learning. For example, presence may play a different role in learning if the goals of the lesson were to recall *what* objects were seen in a room compared to recalling *how to get to* objects seen in a room; the feeling of "being there" may be important in remembering paths between objects relative to the learner's position, but not for remembering features of an object unrelated to the learner's location. Therefore, it is important to determine whether and/or why presence is associated with performance outcomes in different types of learning within a domain, particularly in learning environments with spatial knowledge goals.

1.2. The spatial learning domain: A special place for presence

Many learning and training contexts frequently involve or require spatial understanding; surgery simulation, memory palaces, understanding the inside of a biological cell, or navigation skills training are all examples. Presence may play a particularly strong mediating role in

the spatial learning components of such contexts. Spatial learning involves the integration of visual scanning of a space, mental representation of the relative positions of objects in the space, and mental representation of the relative position of one's body in the space (Hong, Hwang, Tai, Tsai, & 2018). Higher immersion has shown relatively consistent boosts for spatial learning (e.g., Aoki, Oman, Buckland, & Natapoff, 2008; Murcia-López & Steed, 2016; Waller et al., 1998; Zanbaka, Lok, Babu, Ulinski, & Hodges, 2005). For example, Aoki et al. (2008) compared orientation and navigation performance in a space station emergency egress training using HMDs or desktop systems. Trainees using HMDs showed faster pointing times in locating the starting location from a different location or vice versa than those who used desktops. In another example, Zanbaka et al. (2005) found that participants who navigated through a virtual room using a high immersion HMD performed better on map drawing tasks than those who navigated using a low immersion HMD or desktop monitor. The ability to create a spatial mental map of specific landmarks in an environment may benefit more from high immersion than lower immersion. Note that both these examples (map drawing and bearings estimation) involve the most complex and cognitively difficult level of spatial learning, *survey level* knowledge. Presence may be particularly key to this type of spatial learning.

In navigational research, spatial knowledge has been broken down into three general categories per the "landmark-route-survey" (LRS) model (Siegel & White, 1975; Thorndyke & Hayes-Roth, 1982; Lapeyre, Hourlier, Servantie, N'Kaoua, & Sauzeon, 2011). Landmark knowledge is a general understanding of what items were in the environment; route knowledge is an understanding of how to get from one point or item to another; survey knowledge is an overall understanding of the relationship of items in the environment and where they are located in relationship to each other and to one's self (Lapeyre, Hourlier, Servantie, N'Kaoua, & Sauzéon, 2011; Siegel & White, 1975). Landmark knowledge requires the acquisition of semantic and/or sensory information, storage of the representation in long term memory, and the retrieval of the memory when prompted. Route knowledge requires the acquisition, storage, and retrieval of pathways between two landmarks or positions in an environment. Survey knowledge requires the acquisition, storage, and retrieval of landmarks and/or routes from long term memory, as well as the mental manipulation of the direction or orientation of landmarks, routes, or positions within the environment relative to one's position, which is dependent on working memory.

As these different categories of spatial knowledge differ in cognitive mechanisms and cognitive complexity, with landmark knowledge being the simplest and survey knowledge being the most complex, it is possible that presence and levels of immersion affect these categories differently. For example, as survey-level knowledge is more complex than landmark-level knowledge, the acquisition of survey information likely requires higher essential cognitive load to process, and the manipulation of that information in working memory to integrate with prior knowledge and store in long term memory requires more generative cognitive load than landmark information. As the inherent essential and generative cognitive load increase, the mitigation of extraneous cognitive load becomes more important as to not cause *cognitive overload*, and in turn hurt learning. Presence could alleviate extraneous cognitive load by making interactions with the interface more natural. This may work similarly for landmark and/or route knowledge. However, because the acquisition of landmark and route knowledge requires fewer cognitive resources for essential and generative load, the addition of extraneous cognitive load may not cause as much cognitive overload for landmark or route knowledge than survey knowledge. Therefore, presence may play more of a mediating role in survey-level knowledge than landmark-level or route-level knowledge.

1.3. Current study

This study uses a within-subjects design and three levels of immersion to characterize the relationships among immersion, presence, learning outcomes, and types of learning tasks in the spatial domain. Based on the theory of cognitive load, we predict that presence may facilitate learning and may mediate the relationship between immersion and spatial learning, particularly at more difficult higher levels of spatial learning. We found that presence mediated the effect of immersion on spatial learning outcomes, but only for the highest level of spatial learning (survey knowledge), suggesting that the benefits of presence may not emerge unless the nature of the learning task induces sufficient cognitive load. An understanding of this relationship is important for deciding when to use more highly immersive technology for learning.

2. Method

2.1. Participants

Seventy-three participants (50 female, 23 male, ages 19–33 years) were recruited for a 3-session experiment at a southern California university; participants included university students and community members. The voluntary, fully informed, written consent of participants in this research was obtained as required by Title 32, Part 219 of the CFR and Army Regulation 70-25. All human subjects testing was approved by the Institutional Review Board of the U.S. Army Research Laboratory. In a repeated-measures design over three sessions, participants completed spatial learning tasks in three levels of immersive technology with at least 14 days between each session, using an entirely new but similarly difficult virtual environment each time. Sixty-one participants (43 female, 18 male, ages 19–29 years) completed all three sessions of the experiment. Inclusion criteria included being at least 18 years of age and having normal hearing, which was tested using a MAICO MA 40 portable audiometer; normal color vision, which was tested using the 14-plate Ishihara's Test for Color Deficiency; and normal visual acuity (without glasses; contacts were allowed), which was tested using a traditional Snellen acuity chart. Participants were also excluded if they reported contraindicated conditions, including high motion sickness susceptibility, alcohol influence, illness, or pregnancy.

2.2. Materials

2.2.1. Virtual environments

Virtual environments were created using Unity 3D with support scripts in Python. The virtual environments included a controls-familiarization environment, three small task-familiarization (TF) environments used for practice, and three large environments.

The controls-familiarization environment included a plain 3D space with a grid containing squares and spheres. The participant was asked to use the game controller (and HMD motions if applicable) to view, move, and select objects in the familiarization space.

The TF environments were small, themed indoor environments containing theme-relevant objects. The three themed environments were a history museum, recreation center, and holiday rooms. All the TF environments were the same size, and all had four rooms and three doorways. The layout and room connectivities were different for each. The TF environments were used for practicing a scavenger hunt similar to the ones that would take place in the large task environments and for familiarizing participants with the post scavenger hunt environment questions (See Performance tasks). Participants were given 5 min to locate four numbered objects in order, which were marked by a numbered flag. If they finished before the time was up, they were instructed to explore for the remainder of the time. Numbers were used instead of object names to ensure that the objects were encoded based on memory of the virtual experience rather than semantic information given.

Similar to the TF environments, the large environments were also themed indoor environments, which included theme-relevant objects

and contained 13 rooms and 14 doorways. The environments were the same size but had rooms and doorways in different configurations. The environment themes included a home, office, and school. Fig. 1 shows the office environment. Each environment had eight scavenger hunt items distributed to ensure that participants encountered each room in the environment. Scavenger hunts and environment designs were carefully difficulty-balanced across the environments (Files et al., 2019), as were the performance tasks (Sinatra et al., 2019). In the large environment, the participant's task was to complete a scavenger hunt of eight items in the environment, similar to the practice scavenger hunt in the TF environment. Participants were prompted with an instruction to find each object numbered 1 through 8 serially, marked with a numbered flag. During the scavenger hunt, all flags were apparent and disappeared once the hunt was over. This enabled participants to pass by objects they would need to find later. Participants were given 15 min to complete the task and were told to freely explore the environment with the remaining time. Scavenger hunt items were carefully selected so that they were not duplicated across the environments (Sinatra et al., 2019). The environment layout and placement of scavenger hunt items were intended to keep navigational complexity and task difficulty equivalent across all environments (Files et al., 2019).

2.2.2. Presence questionnaires

Participants completed the Presence Questionnaire (PQ, version 4.0) developed by Witmer and Singer (1994, 1998; Witmer, Jerome, & Singer, 2005) to measure presence. It contained 29 items divided into 4 subscales, including the Adaptation/Immersion subscale (8 items), Involvement subscale (12 items), Sensory Fidelity subscale (6 items), and Interface Quality subscale (3 items). Each item was on a 7-point Likert-type scale, and each subscale had relatively good internal reliability: Adaptation/Immersion ($\alpha = .80$), Involvement ($\alpha = .89$), Sensory Fidelity ($\alpha = .84$), Interface Quality ($\alpha = .57$; Witmer et al., 2005). The total PQ score ($\alpha = .91$) was calculated by averaging each item rating (with reversed scored items when appropriate; Witmer and Singer, 1994). Although the internal consistency of the Interface Quality was lower than 0.70, no items were removed from the subscale as this procedure would likely lower the reliability of the total PQ score (Witmer et al., 2005).

Fig. 1. Top-down (top) and first-person (bottom) views of the office environment.

2.2.3. Performance tasks

After each of the environments, participants demonstrated their learning by answering a set of questions pertaining to the environment they just experienced. These tasks were designed to measure three components of spatial knowledge: (1) landmark knowledge, (2) route knowledge, and (3) survey knowledge. The landmark knowledge tasks included a recall task, a yes/no object recognition task, and a multiple-choice object discrimination task. Route knowledge was measured using a route description task, and survey knowledge was measured with a bearings estimation task. Landmark knowledge generally requires less understanding/processing than the more deeply processed route and survey knowledge; route knowledge may require more processing of locations of objects, while survey knowledge requires a full understanding of the relationships between items in the environment (Siegel & White, 1975). Participants completed the performance tasks once after each TF and again after the large environment. Participants who completed all three sessions completed six sets of performance tasks. For the purposes of this manuscript we looked at the performance task sets from all three large environments.

2.2.3.1. Landmark knowledge tasks. First, in the recall task, participants were given the instructions: "Please list all of the objects (target and non-target) you encountered in the virtual environment," and were given 3 min to respond. The task was scored as the number of correct objects recalled. In the yes/no object recognition task, which included 10 questions, the participant indicated whether an object shown was in the environment or not. The yes/no task was scored as the proportion of correct answers. In the multiple-choice object discrimination task, the participant chose an object that was present in the environment from an array of images that included 1 correct answer and 3 foil choices (e.g., 1 correct target clock with 3 similar clocks). The images used for the foil answers were carefully selected to be 3D models of similar quality as the objects used in the environment (Sinatra et al., 2019). The multiple-choice task included 8 questions, and was scored as the proportion of correct answers out of the total number of answered questions. Recall tasks, yes/no objection recognition tasks, multiple-choice object discrimination tasks have previously been used to assess landmark-level spatial knowledge (e.g., Boccia et al., 2017; Kraemer et al., 2017; Nys, Gyselinck, Orriols, & Hickmann, 2015; Zhong & Moffat, 2016). A composite landmark knowledge score was created by averaging standardized scores on the recall, yes/no, and multiple-choice tasks. Fig. 2 shows examples from the yes/no and multiple-choice tasks.

2.2.3.2. Route knowledge task. For the route description task, participants were asked to describe the route from one object to another. The task included 3 questions, and they were given the instructions: "Imagine that you are giving directions to a person who has never been in the environment before. Please provide instructions on how to get from the first item to the second item assuming that your starting place is facing a doorway." The route descriptions were subjectively rated by two post-baccalaureate research assistants with degrees in psychology using a previously developed rubric for route quality based on completeness,

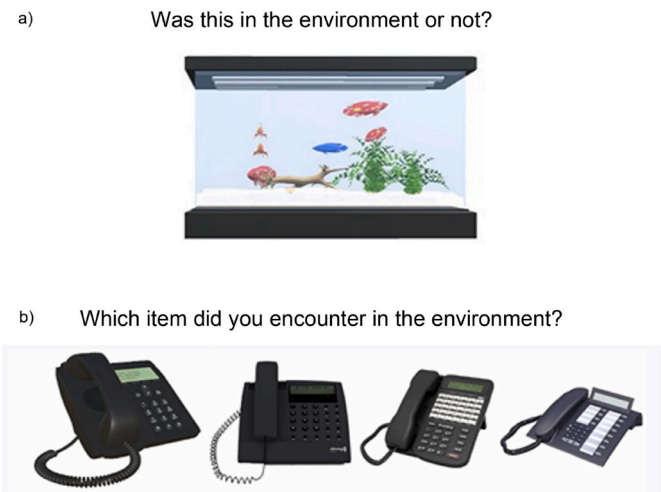

Fig. 2. Sample questions from the a) yes/no task and b) multiple-choice task, measuring landmark knowledge. Figure adapted from Sinatra et al. (2019).

accuracy, and efficiency; the rubric included a 5-point scale, with a 1 indicating bad route quality and a 5 indicating excellent route quality (Lovelace, Hegarty, & Montello, 1999). The raters had good inter-rater reliability (Cohen's quadratic weight kappa = .69, 95% CI [0.64, 0.74]). Route score was calculated as an average of route quality ratings across the 3 questions.

2.2.3.3. Survey knowledge task. For the bearings estimation task, participants were instructed to imagine standing on an object in the environment facing another object and to indicate the direction where a third object was located. They used a circle with tick marks at every 6-degree interval to mark their response, which was modeled after Ragan, Scerbo, Bacim, and Bowman (2017) and implemented using Qualtrics (Oiknine, Files, & Pollard, 2019, pp. 1–11). The bearings task included 6 questions and was scored as the total absolute value of degrees of error. Fig. 3 shows a sample bearings question.

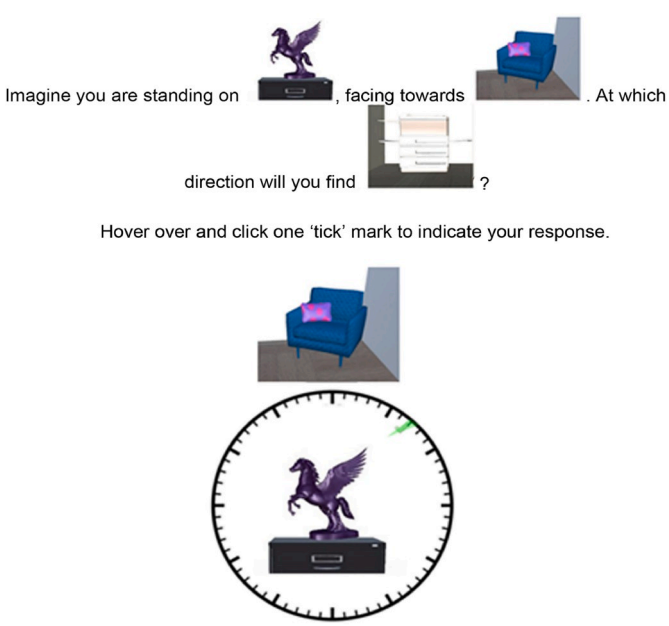

Fig. 3. Sample question from the bearings task, measuring survey knowledge. The green mark indicates a selected answer. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.3. Equipment

The learning environments were displayed in three different levels of immersive technology. The Low immersion condition included a Dell Ultra Sharp 24" Desktop Monitor and Dell OEM AX210 2.0 2-piece USB powered desktop speakers. The Medium immersion condition used a partially occlusive, mid-grade HMD, the NVIS nVisor ST50, fitted with an InterSense InertiaCube4 system, and supra-aural headphones (Sony MDR-G45). The High immersion condition had a fully occlusive HMD, the Oculus Rift CV1, and Audio-Technica ATH-M50x circumaural headphones. The Medium and High immersion HMDs are shown in Fig. 4. All conditions were delivered with an ASUSTek model G752VS gaming laptop running Windows 10 Pro, with NVIDIA Geforce GTX 1070 GPU. In all conditions, participants used an Xbox ONE game controller to walk in the virtual environments. Questionnaires were displayed on a Windows Surface Pro 3 tablet running Windows 10, using Qualtrics software.

Audio playback in the Low, Medium, and High conditions were created using Oculus' 3D audio spatialization effects. The Low condition played only distance-based acoustic intensity cues. The Medium condition played cues based on distance as well as directional head-related transfer function cues with head-tracking. The High condition used distance cues, directional head-motion-tracked sounds, and room acoustic cues, which matched the smooth hard surfaces of the virtual space.

2.4. Procedure

Participants completed the three sessions at least 2 weeks apart (M = 20.08 days). Each participant was tested for normal hearing and vision in their first session. For each session, the participant was set up in his or her assigned level of immersive technology for that session (low, medium, or high). The participant then completed a controller-familiarization task and one of the three TF environments (museum, recreation center, or holiday room), to practice how to move and interact with the objects in the environment. They also completed a small set of practice questions in the same format as the questions after the large environment. They then entered the large environment in one of the three environments (house, office, or school). Following the large environment, the participant completed presence questionnaires and the performance tasks pertaining to the task environment he or she just experienced, which included the recall, yes/no, multiple choice, route description, and bearings estimation tasks. In sessions 2 and 3, the participant completed the other two levels of immersive technology and the other two TF and large environments; each participant experienced each level of immersive technology, TF environment, and large environment only once over the three sessions. The orders of immersive technology, TF environment, and large environment across sessions were counterbalanced across participants.

2.5. Analyses

The independent variable of interest was the level of immersive technology (low, medium, high). Immersive technology was dummy coded with low immersion as the reference category; a planned contrast between medium and high immersion was also run to examine all three comparisons (i.e., low vs. medium, medium vs. high, and low vs. high). Outcome variables included performance on landmark knowledge tasks (averaged from standardized scores of recall, yes/no, and multiple choice tasks), the route knowledge task (average route quality score on a scale from 1 to 5), and the survey knowledge task (total degrees of error; lower scores indicate better performance). An overall composite spatial learning score was calculated by averaging standardized landmark, route, and survey scores, with reverse scored bearings estimation scores to make high scores indicate better performance. PQ score was also used to assess the mediating effects of presence.

Fig. 4. Oculus Rift CV1 HMD used in the high immersion condition (left), and NVIS nVisor ST50 HMD (right) used in the medium immersion condition.

To examine the relationships between level of immersive technology and learning outcomes, immersive technology and subjective presence, and subjective presence and learning outcomes, Linear Mixed Models (LMM) were used. LMMs resolve the non-independence that stems from the repeated measures design (i.e., multiple responses by the same participant) by accounting for the random intercepts of performance on tasks for each individual. The residuals of the outcome variables met assumptions of LMMs; residuals were used to verify homoscedasticity, linearity, and normality (Nobre & Singer, 2007). Restricted Maximum Likelihood (REML) estimation was used with Satterwaithe's method for calculating the effective degrees of freedom, which creates pooled degrees of freedom corresponding to the pooled variances. This method accounts for any differences in variances between groups and produces acceptable Type I error rates even for small sample sizes (Luke, 2017). LMMs were performed using the lme4 (Bates, Maechler, Bolker, & Walker, 2015) and lmerTest (Kuznetsova, Brockhoff, & Christensen, 2017) packages in R (R Core Team, 2019). To examine the mediational pathways of presence, structural equation models were conducted using the lavaan package in R (Rosseel, 2012).

3. Results

3.1. The effect of immersive technology on learning outcomes

Table 1 presents unstandardized parameter estimates of predictors of each learning outcome: landmark knowledge, route knowledge, and survey knowledge. LMMs were conducted using level of immersive technology as a categorical variable. As Table 1 shows, an LMM showed significant differences on landmark knowledge task performance between low and high immersion and between medium and high immersion, but not between low and medium immersion. High immersion led to a 0.36 unit increase over medium immersion and a 0.33 unit increase over low immersion on standardized landmark knowledge score. For route knowledge, there was a marginally significant difference between medium and high immersion on task performance, showing that high immersion led to a score 0.27 units higher than medium immersion. There were no significant differences between low and medium immersion or low and high immersion in route knowledge. For survey knowledge, there was a significant difference in task performance between medium and high immersion, but not between low and medium immersion or low and high immersion. High immersion decreased total degrees of error by 41.43° on the bearings estimation task compared to medium immersion. Immersive technology had a significant effect on overall spatial knowledge using a composite LRS score; the differences between medium and high immersion and between low and high immersion on LRS score were significant, but not between low and medium immersion. High immersion led to better overall spatial performance than medium immersion by 0.34 units and low immersion by 0.20 units.

Table 1
Unstandardized estimates of level of immersive technology as predictors of spatial learning tasks.

Predictor	Estimate	SE	*t*	*p*	95% CI
Landmark Knowledge					
Immersive Technology					
Low vs. Medium	-.03	.10	-.30	.768	(-.24, .17)
Low vs. High	.33	.10	3.22	.002	(.13, .53)
Medium vs. High	.36	.11	3.45	<.001	(.16, .57)
Route Knowledge					
Immersive Technology					
Low vs. Medium	-.14	.14	−1.00	.318	(-.40, .13)
Low vs. High	.13	.13	.98	.327	(-.13, .39)
Medium vs. High	.27	.13	1.97	.051	(.001, .53)
Survey Knowledge					
Immersive Technology					
Low vs. Medium	25.04	16.42	1.53	.130	(-7.22, 57.14)
Low vs. High	−16.40	16.17	−1.01	.312	(-48.04, 15.32)
Medium vs. High	−41.43	16.42	2.52	.013	(9.11, 73.52)
Composite LRS					
Immersive Technology					
Low vs. Medium	-.14	.09	−1.64	.105	(-.31, .03)
Low vs. High	.20	.08	2.36	.020	(.03, .36)
Medium vs. High	.34	.08	3.99	<.001	(.17, .50)

3.2. The effect of immersive technology on presence

To examine whether the level of immersive technology was associated with the learners' subjective presence, LMMs were conducted on PQ scores. Table 2 shows the unstandardized regression coefficients of these analyses. There was a significant difference between medium and high immersion and between low and high immersion, but not between low and medium immersion. High immersion increased PQ presence ratings by 0.37 points compared to low immersion and by 0.46 points

Table 2
Unstandardized estimates of level of immersive technology as a predictor of presence.

Predictor	Estimate	SE	t	p	95% CI
PQ - Total					
Immersive Technology					
Low vs. Medium	-.09	.09	−1.00	.317	(-.28, .09)
Low vs. High	.37	.09	3.97	<.001	(.19, .55)
Medium vs. High	.46	.09	4.91	<.001	(.28, .64)

compared to medium immersion.

3.3. The effect of presence on learning outcomes

To examine whether any of the presence measures had a significant effect on spatial learning outcomes, LMMs were conducted with each outcome variable. PQ scores were significantly associated with performance on route knowledge, survey knowledge, and overall spatial learning, all with higher PQ score indicating better learning performance. As Table 3 shows, an increase in 1 unit of PQ score led to an increase of 0.18 points in route knowledge task performance, a decrease of 25.33 degrees of error in survey knowledge task performance, and an increase of 0.20 units in standardized overall spatial performance.

3.4. Presence as a mediator

The foregoing analyses reveal some relationships between immersive technology, subjective ratings of presence on the PQ, and spatial learning performance. Mediational analyses were also conducted to explore whether presence could explain the effect of immersive technology on spatial learning. That is, does an increase in the level of immersion cause an increase in the learner's presence, which causes better performance on landmark, route, or survey knowledge tasks? Fig. 5 shows unstandardized estimates between immersive technology, presence, and (a) landmark knowledge, (b) route knowledge, (c) survey knowledge, and (d) overall spatial knowledge. For landmark knowledge, presence did not significantly mediate the effect of technology on task performance; the relative indirect effects of immersive technology on landmark knowledge through presence between low and medium immersion, $B = -.000$, $SE = .004$, $z = -0.05$, $p = .961$, medium and high immersion, $B = .002$, $SE = .03$, $z = 0.05$, $p = .961$, and low and high immersion, $B = .002$, $SE = .03$, $z = .05$, $p = .961$, were all non-significant. Presence had similar non-mediating effects with route knowledge. The relative indirect effects between low and medium immersion, $B = .002$, $SE = .01$, $z = 0.19$, $p = .848$, medium and high immersion, $B = .05$, $SE = .04$, $z = 1.17$, $p = .240$, and low and high immersion, $B = .05$, $SE = .04$, $z = -1.21$, $p = .225$, on route knowledge through presence were all non-significant. For survey knowledge, presence significantly mediated the effect of immersive technology on task performance between medium and high immersion, $B = -8.75$, $SE = 4.36$, $z = 2.01$, $p = .045$, and marginally mediated the effect between low and high immersion, $B = -7.64$, $SE = 3.99$, $z = -1.92$, $p = .056$, but presence did not mediate the effect between low and medium immersion, $B = 1.12$, $SE = 1.83$, $z = 0.54$, $p = .541$. Finally, presence did not mediate the effect of immersive technology on overall spatial learning between low and medium immersion, $B = .000$, $SE = .01$, $z = 0.03$, $p = .975$, medium and high immersion, $B = .05$, $SE = .04$, $z = 1.37$, $p = .172$, or low and high immersion, $B = .05$, $SE = .04$, $z = 1.38$, $p = .168$. Thus, presence seems to play a significant role only in mediating performance on survey knowledge tasks, but not on landmark or route knowledge tasks.

Table 3
Unstandardized estimates of presence as predictors of spatial learning tasks.

Predictor	Estimate	SE	t	p	95% CI
Landmark Knowledge					
PQ – Total	.09	.06	1.42	.159	(-.03, .22)
Route Knowledge					
PQ – Total	.18	.08	2.21	.028	(.02, .35)
Survey Knowledge					
PQ – Total	−25.33	9.85	2.57	.011	(5.98, 44.64)
Composite LRS					
PQ – Total	.20	.06	3.43	<.001	(.08, .32)

4. Discussion

4.1. Empirical contributions

The results revealed relationships between immersive technology, presence, and learning outcomes. Overall spatial learning differed between the levels of immersive technology; high immersion led to the best performance compared to medium and low immersion. More specifically, high immersion led to better landmark knowledge and survey knowledge than medium immersion, and better landmark knowledge than low immersion. This corroborates some previous findings that higher immersion leads to higher performance on learning assessments (e.g., Kozhevnikov, et al., 2013; Markowitz et al., 2018; Pollard et al., under review).

The results showed, for all three levels of spatial learning and the composite measure, that the high level of immersion significantly outperformed the medium level but that medium did not significantly outperform the low. The high level of immersion sometimes, but not always, outperformed the low. This yields an overall pattern in which the medium level was generally the worst for spatial learning. Why might this be? Although the immersive properties of the medium and low technologies differed largely (e.g., stereoscopic vs. monoscopic vision, field of view, degrees of head rotation), learning outcomes could not be attributed to these differences. One explanation could be that the medium level of immersion caused cognitive distraction by being a mix of not quite "fake" and not quite "real." The low immersion technology desktop monitor may have been sufficiently fake or abstract (i.e., learners perceived a 2D display) to feel familiar and not be distracting, while the high immersion technology with a high-quality HMD may have been sufficiently realistic (i.e., participants perceived a 3D environment) for participants to again not be distracted by the display media. The medium immersion technology, however, may have been a mix of both as the display may not have been as abstract as the desktop monitor and not as realistic as the high immersion HMD, thereby causing cognitive dissonance in resolving whether it should be perceived as fake or real. A related effect was found in a study by Ragan et al. (2017) examining the effects of amplification of head rotation in virtual environments on target search. Participants were asked to search for objects in a room and complete subsequent pointing tasks using an HMD in which their degrees of head rotation were unamplified (360°) or amplified (270° or 120°), such that smaller physical head rotation corresponded to larger virtual view rotation (e.g., in the 120° condition, turning one's head 120° resulted in viewing 360 degrees of the virtual environment). Participants found significantly fewer objects and had increased orientation error in the 270-degree condition compared to the 360 or 120° conditions. This may be another example of the middle level of immersion being both insufficiently abstract and insufficiently real, which led to worse task performance. These medium levels of immersion may have been associated with the highest level of extraneous load compared to low and high immersion, which could explain poorer performance in those conditions. For low immersion, the desktop monitor could be familiar enough that the interactions with the interface during the learning task were automatized and thereby not distracting. Medium and high immersion both involved a less familiar interface compared to a desktop, but lower resolution displays and slower response times between user actions and responses in the environment in medium immersion conditions may have led to more of a distraction than high immersion. Conversely, higher resolution displays and more naturalistic interactions may have been easier to adapt to, eliminating the cognitive distraction associated with the less familiar immersive technology.

Ratings of presence were the highest in the high immersion technology compared to medium and low immersion, similar to what was found by Cummings and Bailenson (2016). Although the physical properties of the medium immersion technology were more immersive than the low immersion technology, ratings of presence did not differ

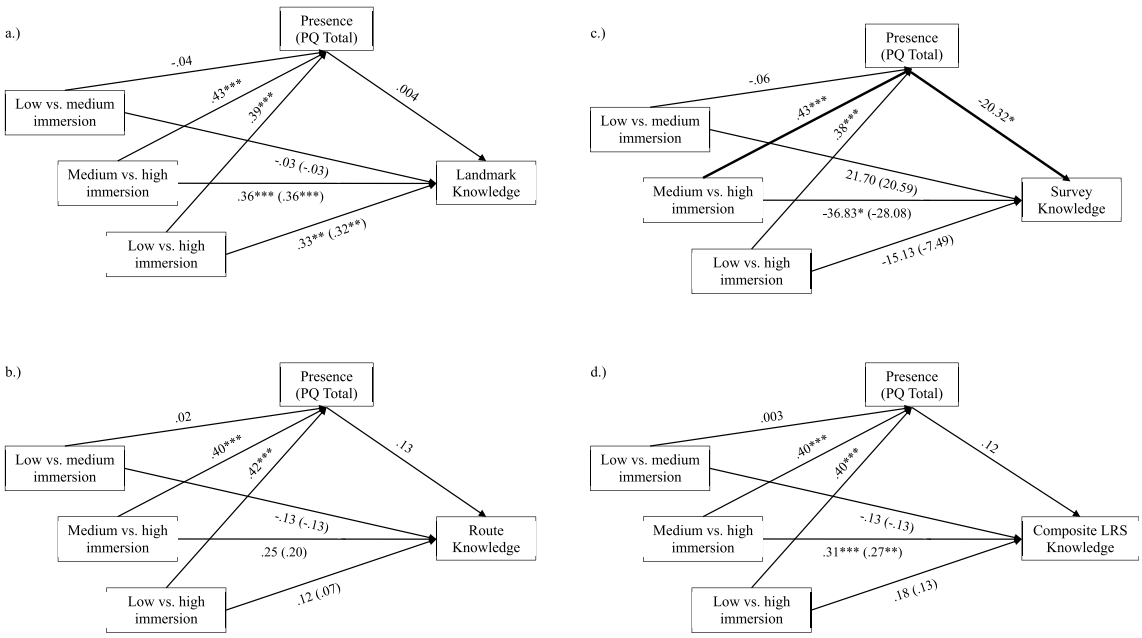

Fig. 5. Mediational models of presence as a mediator of the relationship between immersive technology and (a) landmark knowledge, (b) route knowledge, (c) survey knowledge, and (d) overall spatial task performance, with immersive technology as a 3-level categorical predictor showing all comparisons between low, medium, and high immersion. Unstandardized regression coefficients are shown along the paths. Coefficients in parentheses indicate estimates of the direct effect of immersive technology on knowledge after controlling for presence. Thicker outlines indicate significant indirect effects (i.e., significant mediation). $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

between the two. Again, this could have been due to the quality of the interface between low and medium immersion. Interactions with the desktop interface, although familiar and comfortable, were not immersive, yet interactions with the mid-level HMD may have taken so much effort to adapt to that no more presence was obtained than with the low immersion condition. Perhaps only when a certain level of immersion and ease of adaptation or comfort are obtained together does presence increase.

Ratings of presence were associated with performance on route and survey knowledge, as well as overall spatial knowledge. Some previous research has also found a positive relationship between presence and learning outcomes (e.g., Makransky & Lilleholt, 2018; Schrader & Bastiaens, 2012; Selzer, Gazcon, & Larrea, 2019). Although some previous research and the current study are in line with cognitive theories that can explain the relationships between immersion, presence, and learning, there is also contradicting evidence in which increases in immersion and presence are negatively related to learning outcomes (e.g., Makransky, et al., 2017). However, these relationships were in domains other than spatial learning. More research is needed to elucidate these relationships.

Finally, the results showed that presence significantly mediated the effect of level of immersive technology on survey knowledge task performance, but not landmark knowledge or route knowledge. This suggests that presence may play different roles across different components of spatial knowledge, as well as other non-spatial domains. Presence may not be as important in remembering landmarks or knowing how to get from one landmark to another, but may be crucial in forming a full mental map of an environment. Theoretical explanations are discussed below.

4.2. Theoretical contributions

The finding that presence plays a role in acquiring survey knowledge fits with cognitive load theory (CLT; Chandler & Sweller, 1991) and the cognitive theory of multimedia learning (CTML; Mayer, 2014). The perception of feeling present may have alleviated extraneous cognitive load caused by the physical interface of the HMD and controller, allowing the learner to allocate attention towards learning and processing the relationships between one's self, landmarks, and paths within the environment, and subsequently perform better on survey-level spatial knowledge tasks. Presence may have also helped manage the learner's essential cognitive load during the formation of a coherent mental representation by reducing the cognitive load needed to encode the information. For example, moving through an environment on a computer monitor may require more cognitive resources to translate one's relative position into the environment (i.e., imagining you are in the environment) than an environment in which the user feels present (i.e. already feeling like you are in the environment). Montello, Hegarty, Richardson, and Waller (2004) have also proposed that because virtual environments vary in their physical properties, they provide different information to encode into memory, which may have an effect on what is retrieved later. For example, a desktop display is more abstract than an HMD in that the user has to consciously interpret the spatial information being represented (e.g., translating a 2D image to a 3D first-person view). HMDs offer users a more naturalistic way of acquiring spatial information, which allows users to devote less cognitive resources to this translation. In these types of environments, people have been able to acquire knowledge of an environment in a similar way as they would in the real world (Montello et al, 2004).

Although presence was associated with route knowledge task performance, it did not explain a significant portion of the effect of immersive technology on route knowledge through mediation. This suggests that there may be factors other than presence that could explain this relationships. Survey knowledge may differ from the other types of spatial knowledge in that it is more complex to acquire. It involves remembering landmark locations, understanding routes between them, holding this information in working memory, and integrating this information into a coherent mental map. Because they are less complex than survey knowledge, landmark and route knowledge may not require as much essential cognitive load to acquire or generative load to effectively store in long term memory. Additionally, the process of encoding information about an object or path may not be as affected by

extraneous cognitive load as encoding global information in an environment. Thus, feeling present might not significantly reduce extraneous cognitive load or significantly help manage essential or generative load for better learning. More research is needed to elucidate these differences across types of spatial learning.

4.3. Practical contributions

The results suggest that higher-end HMDs boost complex spatial learning by allowing learners to feel present in the virtual environment. Academic or training lessons that require survey knowledge may benefit from using a highly-immersive HMD and audio system rather than a standard desktop display, medium-immersive HMD, or other less-immersive media. For example, lessons that could benefit from using a highly immersive HMD could be a classroom lecture on understanding the navigational strategies between two parties in WWII or a travel guide on learning how to get around a new city. However, the immersive properties and cognitive effects of mid-level HMDs and mid-level audio warrant further investigation, and practically they may not offer more advantage than a less costly desktop display for many types of learning.

4.4. Limitations and future directions

Although we found effects of immersion and presence within the spatial learning domain, we did not examine their effects in other domains. More research is needed to determine whether these effects are generalizable to training in other domains, including those with spatially relevant information, such as learning the spatial orientation of a molecule in a chemistry lesson, and domains with non-spatially relevant information, such as learning about historic events or people. It may be that only lessons that rely on spatial ability benefit from using immersive technologies. Because of the inconsistency in the relationships between immersive technology, presence, and learning outcomes in previous research, it is important to determine boundary conditions or moderators of the effects of immersive technology, such as domain specificity, on learning and the mediating role of presence. Additionally, presence may not be the only mediating factor that can explain these effects. Future research should examine other factors within the technology or individual that could also play a role.

The sample chosen for the study was a convenience sample from an undergraduate population containing mostly psychology majors, which led to a limited age range and gender disparity and may not generalize to the broader population. However, this study did not aim to study demographic effects, such as gender, on presence and spatial learning. Future research may want to examine differences based on demographic traits by balancing particular traits between groups.

4.5. Conclusions

Overall, this paper contributes to the literature by examining how the subjective feeling of presence mediates the effects of immersive technology on performance and learning. We were able to make a distinction between types of knowledge that was being examined due to the characteristics of the navigation tasks and the existing LRS Model (Siegel & White, 1975). Different spatial domains may have different characteristics within the tasks that should be carefully examined and evaluated. The current work demonstrates that in a spatial task, level of immersion is relevant to learning outcomes. The characteristics of the immersive technology may be related to the ability to form a mental model of the environment; it is possible that when the immersion level is in between low and high, it may take additional processing on the part of the individual to form a mental map of how items in the environment relate to each other. More immersive environments were shown to increase presence, and there was a mediating relationship between level of immersion and a survey knowledge task. Taken as a whole, these results suggest that tasks that require less processing such as landmark and route knowledge may be impacted differently than those that result from deeper processing such as survey knowledge. Careful consideration should be given to the characteristics of the tasks that are being used and the affordances of the immersive technology used to complete them.

Credit author statement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chb.2020.106290.

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